“On the way with Python”

Lecture

05

Python Operators

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Introduction

In Python, operators are special symbols used to perform operations on variables and values. Operators help in performing calculations, comparisons, logical decisions, and data manipulation.

Types of Python **Operator**

- ❑Arithmetic Operators
- ❑Assignment Operators
- ❑Comparison Operators
- ❑Logical Operators
- ❑Bitwise Operators
- ❑Special Operators

Arithmetic Operators

- Used to perform mathematical operations.

Operator	Operation	Example
+	Addition	5 + 2 = 7
-	Subtraction	4 - 2 = 2
*	Multiplication	2 * 3 = 6
/	Division	4 / 2 = 2
//	Floor Division	10 // 3 = 3
%	Modulo	5 % 2 = 1
**	Power	4 ** 2 = 16

Arithmetic Operators

arithmetic_operator.py

```python
a = 7
b = 2

# addition
print ('Sum: ', a + b)

# subtraction
print ('Subtraction: ', a - b)

# multiplication
print ('Multiplication: ', a * b)

# division
print ('Division: ', a / b)

# floor division
print ('Floor Division: ', a // b)

# modulo
print ('Modulo: ', a % b)

# a to the power b
print ('Power: ', a ** b)
```

Assignment Operators

- Used to assign values to variables

Operator	Name	Example
=	Assignment Operator	a = 7
+=	Addition Assignment	a += 1 # a = a + 1
-=	Subtraction Assignment	a -= 3 # a = a - 3
*=	Multiplication Assignment	a *= 4 # a = a * 4
/=	Division Assignment	a /= 3 # a = a / 3
%=	Remainder Assignment	a %= 10 # a = a % 10
**=	Exponent Assignment	a **= 10 # a = a ** 10

Assignment Operators

assignment_operator.py

```python
# assign 10 to a
a = 10

# assign 5 to b
b = 5

# assign the sum of a and b to a
a += b      # a = a + b

print(a)

# Output: 15
```

Comparison Operators

- Used to compare two values. The result is always **True** or **False**.

Operator	Meaning	Example
==	Is Equal To	3 == 5 gives us False
!=	Not Equal To	3 != 5 gives us True
>	Greater Than	3 > 5 gives us False
<	Less Than	3 < 5 gives us True
>=	Greater Than or Equal To	3 >= 5 give us False
<=	Less Than or Equal To	3 <= 5 gives us True

Comparison Operators

comparison_operator.py

```python
a = 5

b = 2

# equal to operator
print('a == b =', a == b)

# not equal to operator
print('a ≠ b =', a ≠ b)

# greater than operator
print('a > b =', a > b)

# less than operator
print('a < b =', a < b)

# greater than or equal to operator
print('a > = b =', a > = b)

# less than or equal to operator
print('a < = b =', a < = b)
```

Logical Operators

- Used to combine conditional statements

Operator	Description	Example
and	True if both are True	True and False → False
or	True if any one is True	True or False → True
not	Reverses the result	not True → False

Logical Operators

```python
x = 10
y = 5
print(x > 5 and y < 10) # Output: True
print(x > 5 or y > 10) # Output: True
print(not(x > 5)) # Output: False
```

Bitwise Operators

- Used to perform operations at the binary (bit) level.

Operator	Description	Example
&	AND	5 & 3 = 1
|	OR	5 | 3 = 7
^	XOR	5 ^ 3 = 6
~	NOT	~5 = -6
<<	Left Shift	5 << 1 = 10
>>	Right Shift	5 >> 1 = 2

Bitwise Operators

```python
a = 5
b = 3

print(a & b)    # Output: 1
print(a | b)    # Output: 7
print(a ^ b)    # Output: 6
print(~a)       # Output: -6
print(a << 1)   # Output: 10
print(a >> 1)   # Output: 2
```

""Thanks All""